

TRAIL INTELLIGENCE, MADE INSPECTABLE

Ask for a loop. *See the ground truth.*

A route planner where a person and an agent work on the same map. The agent discovers the page's own tools, picks a graph-verified circuit, and leaves the evidence on screen.



SATURDAY, 21:40, BARCELONA

“I want to plan a hike with my family, and a trail run with friends, near Barcelona. Two outings — and I don't have much time.

“Googling or asking a chatbot doesn't help much: it's either too generic or not verified. There's Wikiloc, but some trails are old, overgrown with ivy, or plain dangerous. I want something built from official routes.”

OUTING A

Family hike

Short, closed loop. Car access. No retraced legs, no surprises.

OUTING B

Trail run, friends

Longer, more climb, after work. Needs daylight and a shareable brief.

TIME BUDGET:
ABOUT TEN MINUTES, BETWEEN
MEETINGS

WHAT THE THREE OBVIOUS OPTIONS DO

Each one answers a different question than the one asked.

01

Google, or a chatbot

Returns prose. Names a scenic walk but cannot show that the loop closes without a long retraced leg on the map in front of you.

UNVERIFIABLE

02

Crowd GPX archives

No maintenance contract. A 2016 upload can be overgrown, eroded, fenced off or gone — and nothing in the file says so.

STALE & UNOWNED

03

The Park's own material

Authoritative: a published A-E marked-path network and an active notice feed in Catalan. Not a planner — you cannot ask it for a 7 km loop from a car park.

OFFICIAL, NOT PLANNABLE

04

Komoot & AllTrails

Good round-trip planners with elevation. But no official-network evidence, no agent surface, and no proof a suggested loop doesn't retrace itself on a directed graph.

UNPROVEN CLOSURE

A SHARED SURFACE, NOT A CHAT REPLY

Routes proven on a local graph.

Switchback only offers starts and target lengths its directed graph can actually close. The route card separates the length you asked for from the distance that rendered — a 3.5 km target may come back as a 3.6 km circuit, and it says so.

Climb is sampled from ICGC LiDAR. Marked-path coverage is measured against the Park's published network — a preference weight and a disclosure, never a filter, so an 18% loop and a 41% loop can both close honestly.

66

TRAILPACK TILES,
SOURCED IN MANIFEST

A–E

PARK PUBLIC-USE
NETWORK AS EVIDENCE

O

LOGINS, KEYS OR
HIDDEN BACKENDS

PUBLIC TRANSPORT

Vista Rica parking area on the
edge of the Collserola trail graph.Parking
Casseig de les Aigües, verified 21allera
roughly 10–13 cars; it fills

near and GR-92.

de Rei; the simplest access
to Barcelona.

the trail network starting near

at Can Coll.

return circuit are shown.

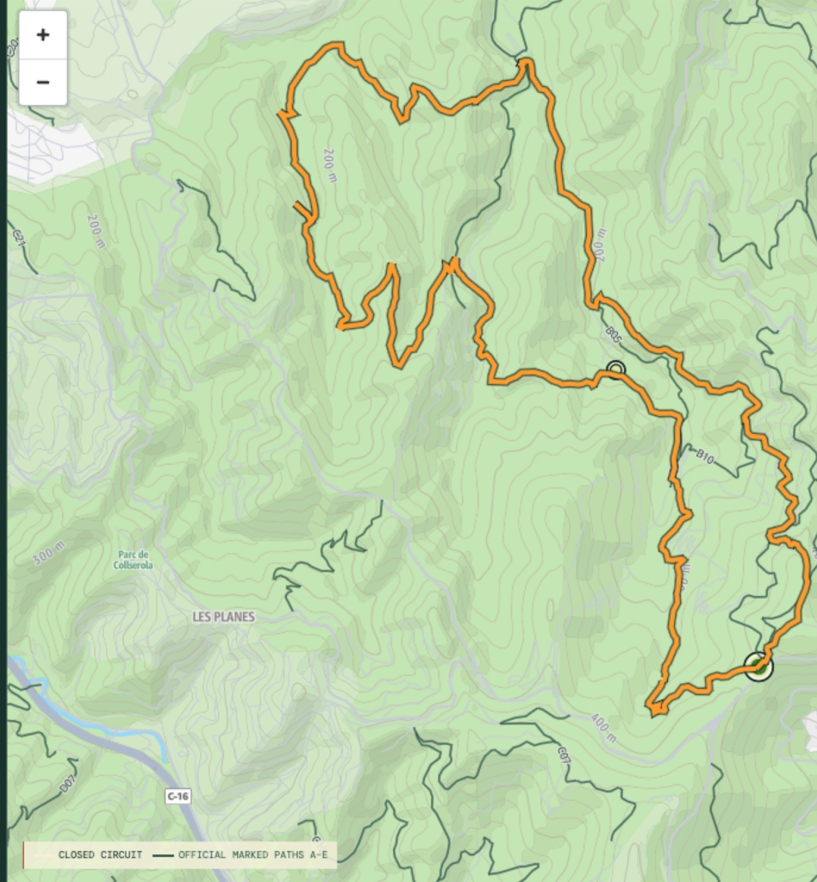
VERIFIED PROFILES

A LOOP
21-
circuit.

14 km.

HARD LOOP
14 km graph-
verified circuit

see the GPX after the loop looks



READY TO WALK

Vista Rica car park TrailPack loop

DISTANCE
14 kmCLIMB
535 mMOVING TIME
3.5 h

Returns to your start · 1% shared access · 18% marked paths · 535 m estimated climb (ICGC LiDAR). Check local conditions before you go.

WHAT “GRAPH-VERIFIED” ACTUALLY MEANS

A directed, weighted trail network.

NODES

Trail junctions, or sampled coordinates along a way — each with a lat/lon.

EDGES

Directed: an allowed travel segment from one node to another. Direction is what makes “non-retracing” checkable.

WEIGHTS

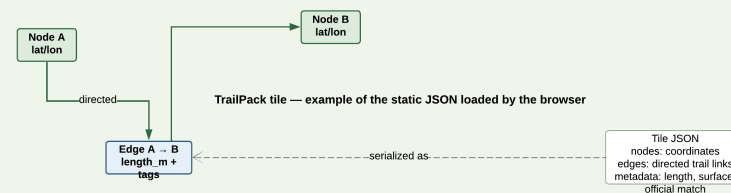
Primarily segment length in metres — which is why a 7 km ask can be answered in 0.5 km steps.

METADATA

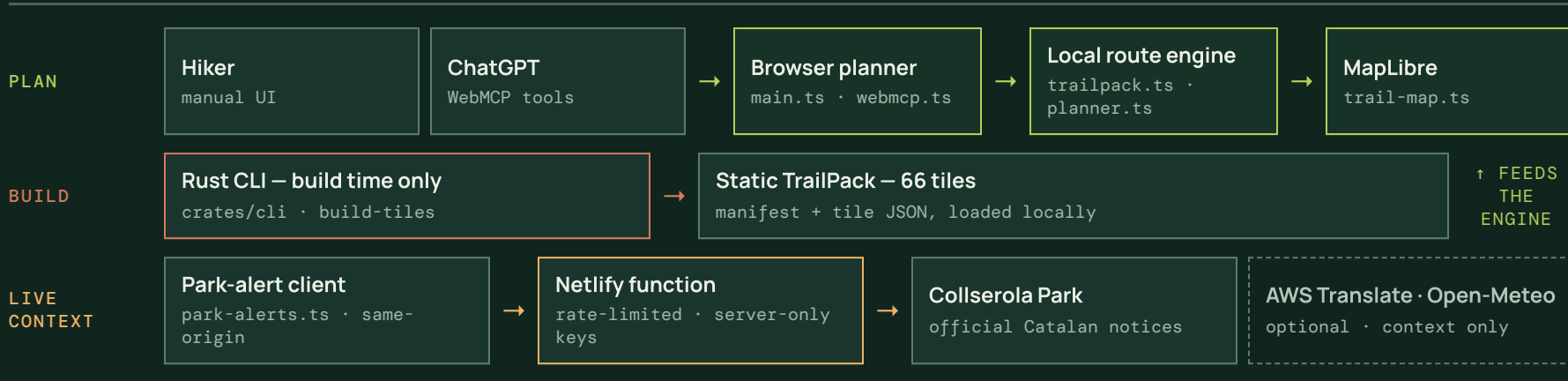
Surface, official A-E route match, and whatever other OSM tags exist — carried on the edge, exposed as evidence.

The same structure used for road networks, built for footpaths. Switchback runs shortest-path and loop-selection over it to construct a non-retracing circuit.

TRAILPACK TILE · THE STATIC JSON THE BROWSER LOADS



66 tiles, built once by a Rust CLI, then routed entirely in the browser. Map tiles are a visual reference layer — not a routing dependency.



Routing stays in the tab

Static TrailPack JSON, read locally. No planning API, no round-trip.

Two entry points, one state

Planner and agent tools drive the same UI and the same route object.

One server-side hop

Only the Park-notice proxy. Credentials never reach the browser.

SHARED TRAIL LEDGER

Nothing happens off-map.

The agent searches the graph. The person keeps every irreversible act. Both are looking at the same route state, and every call is on the record.

HUMAN

Asks in plain language

"A 7 km family loop from Vista Rica, prefer marked paths — and when in the next three days?"

AGENT

Discovers the page tools

`list_circuit_options`, then a dry run with `validate_circuit` — before anything is drawn.

AGENT

Draws it on your map

`plan_route` renders the loop *before* it returns, and pulls forecast, daylight and Park notices with it.

HUMAN

Inspects and pushes back

User can talk to agent about the idea - choose different path

HUMAN

Sends it, or doesn't

The agent prepares the group brief and the GPX. Only a human click shares the message or starts the download.

GETTING THERE

BY CAR

PUBLIC TRANSPORT

CHOOSE A START

Vista Rica car park

Car-access trailhead at the Vista Rica parking area on the Arrabassada road, verified 14 m from the Collserola trail graph.

Passeig de les Aigües parking

Your suggested parking start at Passeig de les Aigües, verified 21 m from the Collserola trail graph.

Parking Itineraris La Budellera

Revolt de les Monges parking for roughly 10–13 cars; it fills quickly on fair weekends.

Pantà de Vallvidrera

Roadside parking near the reservoir and GR-92.

Santa Creu d'Olorda

Western car approach from Molins de Rei; the simplest access from Tarragona without crossing Barcelona.

Can Jané, Sant Cugat

Car access for Sant Medir, with the trail network starting near Can Jané.

Can Coll, Cerdanyola

Quiet northern-sector car start at Can Coll.

Only starts with a graph-verified return circuit are shown.

CONSTRAINT AS A FEATURE

Fixed options. Not limiting ones.

ONLY VERIFIED STARTS

Seven trailheads appear because the graph can close a return circuit from each. Car or public transport, filtered in one tap.

LENGTHS, NOT DIFFICULTY

Short / medium / long describe how long you'll be out. 2 km for the family, 14 km for the run — from the same car park.

±0.5 KM, STATED

Targets snap to 0.5 km steps and a result is accepted only inside tolerance. The card shows both numbers, never one pretending to be the other.

ESCAPE HATCHES

Prefer waymarked paths, avoid a named segment, or move a waypoint by hand. Each replan re-proves closure or refuses.

TWO USE CASES

Family hike and trail run are not two products — they are two profiles over the same verified network, planned in the same two taps.

FROM ROUTE TO DECISION

The last mile is a group chat.

OFFICIAL PARK ALERTS

Risc d'incendi: nivell 3 al Parc Natural de la Serra de Collserola

2026-08-29 · parcnaturalcollserola.cat

MACHINE TRANSLATION

Fire risk: level 3 in Collserola Natural Park

Read server-side, rate-limited, source-linked.
Switchback does not decide whether a notice applies to your exact route.

SHARE WITH FAMILY OR FRIENDS

Vista Rica loop — Sunday 09:30

7.1 km closed circuit · 310 m climb
~2 h 10 moving · 41% official marked
Start: Vista Rica car park (car)

Forecast: ☀ 09:00-12:00, 19°C,
5% rain, gusts 12 km/h
Sunset 20:44 — plenty of margin

Park notices: 1 active (fire risk level 3) — no open trails closed
Bring water; check signs on the day.

COPY BRIEFING

No message is sent. No messaging account is touched.

Official notices, a message your family will actually read, and a trace your watch can follow. All prepared by the agent; all released by you.

TAKE THE MAP WITH YOU

GPX 1.1, full resolution

Five named TrailPack transitions so the turns mean something on a watch face. Prepared on request, revealed as a control — never auto-downloaded.

DOWNLOAD GPX ↴

PLAN ANOTHER ROUTE

The interactive map stays live in the tab: pan, zoom, tap a marked path to identify it.

REGISTERED WITH THE BROWSER MODEL CONTEXT

Fourteen small tools. One accountable route.

○ READ-ONLY ● MUTATES STATE

DISCOVERY

○ list_circuit_options

○ get_route_summary

VALIDATION

○ validate_circuit

○ explain_difficulty

○ explain_segment

○ describe_last_edit

LIVE CONTEXT

○ get_trail_weather

○ get_park_alerts

● get_hiking_conditions

MUTATION

● plan_route

● avoid_segment

HANDOFF

● prepare_gpx

● prepare_route_briefing

● record_session_note

Every call and its validated input lands in the page's visible invocation log. The manual planner runs the same actions, so the page works with no agent at all — labelled browser-demo mode.

WHAT SWITCHBACK WILL NOT CLAIM

A planner that says “unrated” is worth more than one that guesses.

The graph is not the ground

TrailPack does not prove current signs, closures, surface condition, obstacles, exposure or grade. Check local conditions before you leave.

Marked ≠ waymarked today

A-E preference is evidence from the Park's published network, not a statement about present-day signage on the ground.

Live sources may be missing

If forecast or Park notices fail, the panel says so and the route stays clearly labelled as TrailPack-only evidence. No fabricated fallback.

Never a safety clearance

`get_hiking_conditions` assembles decision context and names what still needs local checking. It does not certify that hiking is safe.

WHY THIS BELONGS ON THE OPEN WEB

Trail planning stops being a text suggestion. *It becomes a shared, auditable decision.*

One family hike, one trail run, ten minutes, no account. The agent did the searching. The map, the notices and every tool call stayed where a person could argue with them.

The pipeline is region-agnostic: any OSM extract plus any official trail network builds a new TrailPack. Collserola is the demo region because the Park publishes a machine-checkable A-E network.

switchback-mvp-
igor.netlify.app ↗

NO ACCOUNT · NO KEY · NO
CREDENTIAL

PASTE THIS INTO A WEBMCP BROWSER

Use the site tools on this Switchback page.

Plan me a long trail run from a car park in
Collserola, with my friends, this Saturday.

Expected: the map becomes a closed loop, the route card fills with distance and sampled ascent, the ledger shows the agent's action, and every live panel states whether its source was reachable.

OR PRESS THE PAGE'S OWN "COPY AGENT TEST PROMPT".